Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “net force” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

01 Warm-up: bring back an earlier idea

A rover travels straight east at a steady 6 m/s. A second rover turns at the same steady speed. Which can have zero net force?
Explain using velocity.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

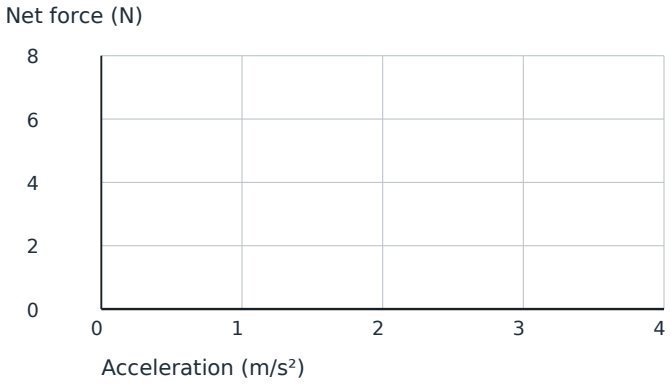
02 Worked example: explain the model

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A 2 kg cart has net forces 2 and 6 N in two trials. Using a=F/m gives accelerations 1 and 3 m/s². Tripling net force triples acceleration.

Your turn: Draw a force-versus-acceleration graph for these trials.
Explain the meaning of the slope.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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Sentence starters are optional. You may plan in any language.

03 Read a proportional pattern

Use the supplied model data for the same 2 kg system. What happens to acceleration when force doubles?
Predict a at 8 N.

Sentence starter: I predict … . My model supports this because … .

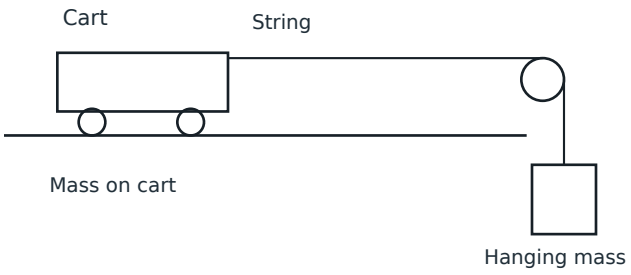
Net force (N)	a(m/s²)
2	1
4	2
6	3

Marzano check - circle: 1 2 3

04 Keep mass constant

A student increases the hanging mass by adding a mass from the supply box.
Use the setup model to identify the two quantities this changes.
Describe how to change driving force while keeping total moving mass constant.

Sentence starter: I notice … . This supports … because … .

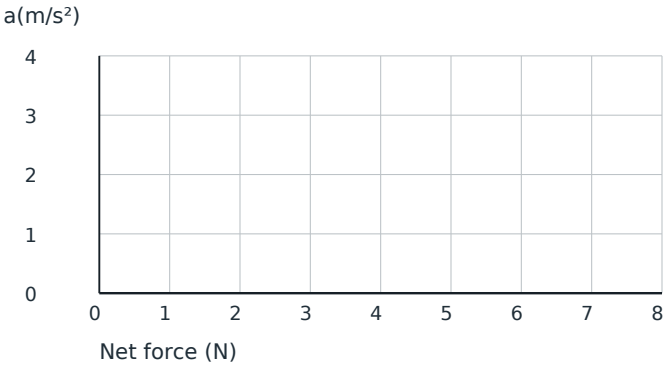


Marzano check - circle: 1 2 3

05 Graph the pattern

Plot the model data from 3.
Label both axes and draw a trend line. If you collect your own data, put those in a separate table and label them measured.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

With a fixed net force, a 3 kg system accelerates at 10 m/s².
Predict acceleration for a 6 kg system.
State a control needed for this comparison.
Use the table to keep the controlled and changed quantities distinct.

Word help: acceleration = rate of change of velocity

Sentence starter: I predict … . My model supports this because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

07 Exit check

A fixed-mass system accelerates at 2 m/s² under 4 N net force.
Predict its acceleration under 6 N net force.
Explain with a comparison.

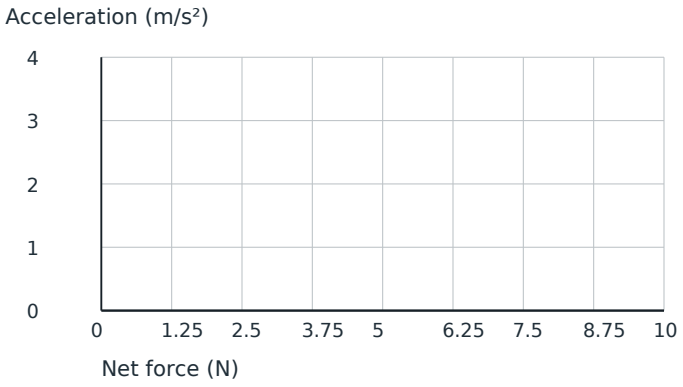
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design three net-force settings that give accelerations 1, 2, 3 m/s² for a 3 kg cart.
Explain what must stay fixed.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: 1 2 3